

Element A

Presentation And Justification Of The Problem

Due to the lack of space outside and the uncertain temperatures in Maryland during the winter of 2024, the use of hydroponics is the only certain way to grow plants in school for clubs and classes. Not only that, but due to germination of seeds in paper towels leading to mold, hydroponics is the best way to germinate seeds without a high risk of mold growth.

Our stakeholder is a biology teacher and the sponsor for the gardening club at Eleanor Roosevelt High School. She requires another hydroponic unit to be built to have enough units for her biology and gardening club students to use, so they can learn how to garden certain plants. The students of the Homegrown Heroes Club have already experimented with hydroponics from sponsors, but due to there being only two units, to get another hydroponics case, one would have to be bought. However, these machines are newer, meaning they are expensive. With the cost problem, students in the Biology class are also unable to learn with the use of the hydroponic gardens due to all the hydroponic units being in use. Again due to the small number of available hydroponic planters, many plants cannot be planted without sacrificing another plant to be taken out. Due to their natural cycle, taking it out of the machine may cause it to react negatively either dying or faltering before getting better. Times like this winter makes the plants extremely susceptible to being in uncommon environments and ones that do not support their needs like the hydroponic machine. The effects of poor treatment was proven in multiple previous studies stating it can cause the plant “stress” and can cause many problems within the plant if its treatment is not to its liking. Plant growth and distribution are affected by the environment. Most plant problems are caused by environmental stress. Factors that affect plant growth include light, temperature, water, humidity, and nutrition. Understanding these factors can help manipulate plants for increased leaf, flower, or fruit production. ([SOURCE](#)) According to a credible article from the University of Minnesota Extension, “Hydroponic growing allows for faster growth and higher yields than traditional soil-based growing systems.” This further strengthens the need of a hydroponic planter in a classroom/club since there is such little time in the school year, let alone after school hours, to grow plants. Not to mention, most of the school year does not have the optimal weather for traditional planting ([SOURCE](#)) . However, with hydroponic planters, an article from Eden Green Technology states they, “can grow crops in any climate.” These factors severely increase the amount of plants the teacher, her biology students, and the gardening club can plant ([SOURCE](#)). Another previous study stated that hydroponics has an amazing impact on indoor gardening and overall convenience for those with smaller land. There are numerous advantages to using hydroponic systems. Firstly, since there is no soil involved, there is no need to worry about having a piece of land, dealing with weeds or pathogens in the soil, or using pesticides on the crops. Additionally, water is conserved as the same water can be reused due to the nutrient reservoir. A blog posted by Tuhopeta reads, “Hydroponic planting can recycle water, which can reduce water waste and achieve higher utilization efficiency than traditional soil planting.” This can have a positive impact on the school

and the earth by conserving their resources ([SOURCE](#)). Furthermore, because most hydroponic farms are located indoors, food can be produced throughout the year, even in densely populated areas like New York City. With all these benefits, it is likely that society will see an increase in the number of hydroponic farms across the US and around the world. This method of farming has the potential to revolutionize agriculture by reducing water usage and other resources. ([SOURCE](#)) Reading from these sources, the team can concur that the main purpose of this hydroponics system is to grow food that is easy to harvest and can regrow after pruning. This design also has to be moveable, easy to clean and overall needs to be easy to control. The hydroponics planted should be around 30 by 60 inches to fit within the desired place in the classroom. In order to assess the functionality of the garden, it is necessary to conduct tests on the seeds as well as evaluate the overall structure to determine its effectiveness and identify any potential design flaws that may result in leaks or other issues. This evaluation process not only benefits the classroom but also provides valuable insights for future gardeners seeking a cost-effective and straightforward method to create a hydroponics garden in their own homes. During our research to find suitable solutions for our specific problem, the team encountered a challenge in locating adequate sources that offered affordable options with clear instructions. As a result, the team had to draw inspiration from various suggestions provided by the hydroponics community in order to effectively address our issue.